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Mark Scheme (Results)

March 2018

Functional Skills Mathematics Level 2

FSM02

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FUNCTIONAL SKILLS (MATHEMATICS)
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Guidance for Marking Functional Skills Maths Papers

General

- All candidates must receive the same treatment. You must mark the first candidate in exactly the same way as you mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- All the marks on the mark scheme are designed to be awarded. You should always award full marks if deserved, i.e. if the answer matches the mark scheme. You should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.

Applying the Mark Scheme

- The mark scheme has a column for **Process** and a column for **Evidence**. In most questions the majority of marks are awarded for the process the candidate uses to reach an answer. The evidence column shows the most likely examples you will see if the candidate gives different evidence for the process, you should award the mark(s).
- **Finding 'the answer'**: in written papers, the demand (question) box should always be checked as candidates often write their 'final' answer or decision there. Some questions require the candidate to give a clear statement of the answer or make a decision, in addition to working. These are always clear in the mark scheme.
- If working is **crossed out and still legible**, then it should be marked, as long as it has not been replaced by alternative work.
- If there is a **choice of methods** shown, then mark the working leading to the answer given in the answer box or working box. If there is no definitive answer then marks should be awarded for the 'lowest' scoring method shown.
- A suspected **misread** may still gain process marks.
- It may be appropriate to **ignore subsequent work (isw)** when the candidate's additional work does not change the meaning of his or her answer.
- You will often see correct working followed by an incorrect decision, showing that the candidate can calculate but does not understand the functional demand of the question. The mark scheme will make clear how to mark these questions.
- **Transcription** errors occur when the candidate presents a correct answer in working, and writes it incorrectly (on the answer line in a written paper); mark the better answer.
- **Incorrect method** if it is clear from the working that the "correct" answer has been obtained from incorrect working, award 0 marks. Send the response to review for your Team Leader to check.
- **Follow through marks (ft)** must only be awarded when explicitly allowed in the mark scheme. Where the process uses the candidate's answer from a previous step, this is clearly shown. Speech marks are used to show that previously incorrect numerical work is being followed through, for example **'240'** means **their** 240.
- Marks can usually be awarded where **units** are not shown. Where units, including money, are required this will be stated explicitly. For example, 5(m) or (£)256.4 indicates that the units do not have to be stated for the mark to be awarded.

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- **Correct money notation** indicates that the answer, in money, must have correct notation to gain the mark. This means that money should be shown as £ or p, with the decimal point correct and 2 decimal places if appropriate.
  e.g. if the question working led to £12 ÷ 5,
  Mark as correct: £2.40 240p £2.40p 2.40£ Mark as incorrect: £2.4 2.40p £240p 2.4 2.40 240
- Candidates may present their answers or working in many **equivalent** ways. This is denoted **oe** in the mark scheme. Repeated addition for multiplication and repeated subtraction for division are common alternative approaches. The mark scheme will specify the minimum required to award these marks.
- A **range** of answers is often allowed:
  - [12.5, 105] is the inclusive closed interval
- **Parts of questions:** because most FS questions are unstructured and open, you should be prepared to award marks for answers seen in other parts of a question, even if not explicit in the expected part. E.g. checks in on earlier answer box.
- **Graphs**
  The mark schemes for most graph questions have this structure:

Process	Mark	Evidence
Appropriate graph or chart – (e.g. bar, stick, line graph)	1 or	1 of: linear scale(s), labels, accurate plotting (2 mm tolerance)
	2 or	2 of: linear scale(s), labels, accurate plotting (2 mm tolerance)
	3	all of: linear scale(s), labels, accurate plotting (2 mm tolerance)

The mark scheme will explain what is appropriate for the data being plotted.
A **linear scale** must be linear **in the range where data is plotted,** and use consistent intervals. The scale may not start at 0 and not all intervals must be labelled. Thus a graph that is 'fit for purpose' is one where the **data is displayed clearly and values can be read**, will gain credit.
The minimum requirements for **labels** will be given, but you should give credit if a title is given which makes the label obvious.
Plotting must be correct for the candidate's scale. Candidate's scale must be in numerical order. Award the mark for plotting if you can read the values, even if the scale is not linear.
The mark schemes for **Data Collection and/ or summary Sheets** refer to **input opportunities** and to **efficient input opportunities.** When a candidate gives an input opportunity, it is likely to be an empty cell in a table, it may be an instruction to 'circle your choice', or it may require writing in the data in words. These become efficient, for example, if there is a well-structured 2-way table, or the input is a tick or a tally rather than a written list.

Discuss any queries with your Team Leader.

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Section A: Fitness club

Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q1	I6	Identifies values from the graph	1	A	3 of 166, 158, 148, 146 and 144 seen
	A4	Process to work with mean	1 or	B	('166' + '158' + '148' + '146' + '144') ÷ 5 (= 152.4) **OR** '166' + '158' + '148' + '146' + '144' (=762) **and** 150 × 5 (=750)
	I7	Valid decision with accurate figures	2	BC	Yes **AND** 152.4 **OR** Yes **AND** 762 **and** 750 Accept Yes **AND** 152 with no error seen.
	A5	Valid check	1	D	Valid check, e.g. reverse calculation or alternative method
		Total marks for question	**4**		

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Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q2(a)	R1	Begins process to work with fraction or percentage	1 or	E	499 ÷ 8 (= 62.375) oe **OR** 369.59 ÷ 100 × 20 (= 73.918) oe
	R3	Full process to find one cost after decrease or increase or engages with both fraction and percentage	2 or	EF	499 ÷ 8 × 7 (= 436.625) oe **OR** 369.59 ÷ 100 × 120 (= 443.508) oe **OR** 499 ÷ 8 (= 62.375) oe **and** 369.59 ÷ 100 × 20 (= 73.918) oe
	A4	Full process to find figures to compare	3 or	EFG	499 ÷ 8 × 7 (=436.625)oe **and** 369.59 ÷ 100 × 120 (=443.508) oe
	I7	Valid decision with accurate figures	4	EFGH	Best Bikes **AND** (£)[436, 437] **and** (£)[443, 444] **OR** Best bikes **AND** (£)6.8(75) (difference)

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Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q2(b)	R1	Engages with the problem	1 or	J	e.g. 162 – 135 (= 27) **OR** 135 ÷ 162 (= $\frac{5}{6}$**)** **OR** 162 ÷ 5 (= 32.4) oe **OR** 135 ÷ 4 (= 33.75)
	A4	Full process to find figures to compare	2 or	JK	e.g. 162 – 135 (= 27) **and** 162 ÷ 5 (= 32.4) **OR** $\frac{'27'}{162}\left(=\frac{1}{6}\right)$ oe **OR** $1-\frac{'5'}{6}\left(=\frac{1}{6}\right)$ oe **OR** '27' ÷ 162 × 100 (= 16.6..) **(and** 1 ÷ 5 × 100 (= 20**)** oe) **OR** 162 – '32.4' (= 129.6) oe **OR** 135 + '32.4' (= 167.4) **OR** '33.75' × 5 (= 168.75)
	I7	Valid decision with accurate and consistent figures	3	JKL	e.g. No **AND** 27 **and** 32.4 (beats per minute difference) **OR** No **AND** $\frac{1}{6}$ **OR** No **AND** [16, 17] (%) **and** 20(%) **OR** No **AND** 0.1(6..) **and** 0.2 **OR** No **AND** 129.6 (beats per minute) **OR** No **AND** 167.4 (beats per minute) **OR** No **AND** 168.75 (starting beats per minute)
		Total marks for question	**7**		

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Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q3(a)	R1	Process to work with probability	1 or	M	100 – 62 (= 38) oe
	A4	Correct answer in correct probability notation	2	MN	38% oe
Q3(b)	R2	Begins to work with 60 seconds in a minute	1	P	e.g. 60 – 22 (= 38) **OR** 22 + (38) = 60 (build up to 60) **OR** 14 (mins) (and) 60 (secs) **OR** 60 × 15 (= 900) **or** 60 × 13 (= 780) **OR** (60 × 15) + (60 × 60 × 2) (= 8100 secs) **OR** (60 × 13) + (60 × 60 × 2) + 22 (= 8002 secs) May be seen on a diagram
	R1	Valid process to find the difference	1 or	Q	e.g. '38' + 60 (= 98) **OR** 13 (mins) 22 (secs) + '38'(secs) + 1 (min) (= 15 mins) oe **OR** '135' (mins) – '133 (mins) 22' (secs) (= 1 min 38 secs) oe **OR** '8100' – '8002' (= 98)
	I6	Accurate figure with units	2	QR	98 secs **OR** 1 min 38 seconds **or** 1.63(3..) mins NB If awarding R award P
Total marks for question			**5**		

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Section B: The house

Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q4(a)	R3	Begins to engage with a decimal amounts of monthly rent	1 or	A	895 × 0.7 (= 626.5) oe **OR** 895 × 1.25 (= 1118.75) oe **OR** 1 + 0.7 (= 1.7) **or** 1 + 1.25 (= 2.25) **or** 1.25 + 0.7 (= 1.95) oe
	A4	Develops solution	2	AB	895 × 0.7 (= 626.5) oe **and** 895 × 1.25 (= 1118.75) oe **OR** 895 + '626.5' (= 1521.5) **OR** 895 + '1118.75' (= 2013.75) **OR** 1 + 0.7 + 1.25 (= 2.95) oe
	I6	Full process to work out total cost or total of share	1	C	895 + '626.5' + '1118.75' (= 2640.25) **OR** 895 × '2.95' (= 2640.25) **OR** '537' + '375.9'+ '671.25' (= 1584.15)
	R2	Begins process to work with ratio	1 or	D	'2640.25' ÷ (3+2) (= 528.05) **OR** 895 ÷ (3+2) (= 179) **OR** '626.5' ÷ (3+2) (= 125.3) **OR** '1118.75' ÷ (3+2) (= 223.75)
	A4	Full process to work with ratio	2	DE	'528.05' × 3 (= 1584.15) **OR** '179'× 3 (= 537) **OR** '125.3' × 3 (= 375.9) **OR** '223.75' × 3 (= 671.25)
	I6	Accurate figure in correct money notation	1	F	£1584.15 If F mark awarded, award all 6 marks. NB 1584.15 without a pound sign should be awarded 5 marks.

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Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q4(b)	R1	Begins to work with formula	1 or	G	e.g. 3000 × 1.034 (= 3102) **OR** 3000 ÷ 18 (= 166.66..) **or** 1.034 ÷ 18 (= 0.0574..) **OR** 170 × 18 (= 3060)
	A4	Full substitution into formula or finds figures to compare	2 or	GH	e.g. '3102' ÷ 18 (= 172.33..) **OR** '166.66' × 1.034 (= 172.33..) **OR** '0.0574..' × 3000 (=172.33..) **OR** '3102' ÷ 170 (= 18.24..) **OR** 3060 ÷ 1.034 (= 2959.38) **OR** 3000 × 1.034 (= 3102) **and** 170 × 18 (= 3060)
	I7	Valid decision with accurate figures	3	GHJ	Yes **AND** (£)[172.3, 173] **OR** Yes **AND** 18.2(4..) (months) **OR** Yes **AND** (£)2959(.38**) OR** Yes **AND** (£)3102 **and** (£)3060
	A5	Valid check	1	K	Valid check, e.g. reverse calculation or alternative method
		Total marks for question	**10**		

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Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q5(a)	A4	Begins to work with perimeter or finds missing length	1 or	L	e.g. 3.5 + 2 + 4.7 (= 10.2) **OR** 2 + 1.25 (= 3.25) **or** 4.7 – 3.5 (= 1.2)
	I6	Full process to find total perimeter	2	LM	(4.7 + 3.25) × 2 (= 15.9) oe May be seen in subsequent working
	A4	Process to engage with the multiplier	1 or	N	e.g. '15.9' × 1.5 (= 23.85) oe **OR** 25 ÷ 1.5 (= 16.66..) **OR** 25 ÷ '15.9' (= 1.572..)
	I7	Valid decision with accurate figures	2	NP	Yes **AND** 23.85(m) **OR** Yes **AND** 15.9(m) **and** 16.6(6..)(m) **OR** Yes **AND** 1.57(2..) (times around) NB If NP awarded, award all 4 marks
Q5(b)	R2	Begins to work with scale	1 or	Q	Rectangle with one side 7 sq **or** 5 sq **OR** Rectangle with sides in the ratio 7:5 **and** long side against a wall **and** 1 sq away from the door, the window, the shelves and fireplace **OR** 175 ÷ 25 (= 7) **or** 125 ÷ 25 (= 5) **or** 25 ÷ 25 (= 1) May be indicated on diagram
	I6	Fully correct solution	2	QR	Rectangle 7 sq by 5 sq **and** long side against a wall **and** 1 sq away from the door, the window, the shelves and fireplace
		Total marks for question	**6**		

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Section C: Limousine hire

Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q6	R2	Process to find one of the missing lengths	1	A	8.2 – 4.3 (= 3.9) **OR** 6.5 – 3.1 (= 3.4)
	I6	Process to find one relevant area	1 or	B	e.g. 8.2 × 6.5 (= 53.3) **or** 8.2 × 3.1 (= 25.42) **or** 4.3 × 6.5 (= 27.95) **or** 0.5 × '3.9' × '3.4' (= 6.63) **or** 4.3 × '3.4' (= 14.62) **or** '3.9' × 3.1 (= 12.09)
	A4	Full process to work out total area or total number of packs required if individual areas used	2	BC	e.g. '53.3' – '6.63' (= 46.67) **OR** '25.42' + '14.62' + '6.63' (= 46.67) **OR** '27.95' + '12.09' + '6.63' (= 46.67) **OR** '6.19..' + '2.68..' + '1.47..' (= 10.34..)
	A4	Process to calculate number of packs for total area or partial area	1	D	e.g.'46.67' ÷ 4.51 (= 10.34..) **OR** '27.95' ÷ 4.51 (= 6.19..) **or** '12.09' ÷ 4.51 (= 2.68..) **or** '6.63' ÷ 4.51 (= 1.47..)
	I6	Functionally rounded figure for number of packs	1	E	11 (packs)
Total marks for question			**5**		

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Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q7	R1	Begins to work with constraints	1 or	F	2 customers placed correctly with gaps and start times **OR** 3 customers placed correctly without gaps and start times
	R2	Develops solution	2 or	FG	3 customers placed correctly with gaps and start times **OR** 4 customers placed correctly without gaps and start times
	I7	Fully correct solution	3	FGH	4 customers placed correctly with clear start and finish times with at least 45 minute gap between bookings
Total marks for question			**3**		

Driver: Adnan	**Day:** Saturday
Time	**Client**
9 am to 12 pm	Magenty family
1 pm to 2.30 pm	Ms S(ingh)

Driver: Fayez	**Day:** Saturday
Time	**Client**
9 am to 12 pm	Mr H(arding)
1 pm to 3 pm	Mr and Dr Pacitti

Driver: Toni	**Day:** Saturday
Time	**Client**
9.30 am to 11 am	B(arker) family
12 pm to 1.30 pm	Mr and Mrs K(han)
2.30 pm to 5 pm	Lee family

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Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q8	I6	Works with mileage	1	J	2834 – 2529 (= 305)
	A4	Works with proportion	1 or	K	e.g. 75.2 ÷ 4.54 (= 16.56..) **OR** '305' ÷ 20 (= 15.25) **OR** 75.2 ÷ '305' (= 0.24..) **or** 4.54 ÷ 20 (=0.227) **OR** '305' ÷ 75.2 (= 4.055..) **or** 20 ÷ 4.54 (= 4.405..)
	R3	Full process to find figures to compare	2 or	KL	e.g. '305' ÷ '16.56..' (= 18.41..) oe **OR** 75.2 ÷ 4.54 (= 16.56..) **and** '305' ÷ 20 (= 15.25) **OR** '0.24..' × 20 (= 4.93..) **OR** '16.56..' × 20 (= 331.27..) **OR** '15.25' × 4.54 (= 69.235) **OR** 75.2 ÷ '305' (= 0.24..) **and** 4.54 ÷ 20 (=0.227) **OR** '305' ÷ 75.2 (= 4.055..) **and** 20 ÷ 4.54 (= 4.405..)
	I7	Valid decision with accurate figures	3	KLM	e.g. No **AND** [18.4, 18.5] (mpg) **OR** No **AND** [16.5, 16.6] **and** 15.25 (gallons) **OR** No **AND** 4.9(3..) (l) **OR** No **AND** 331(.27..) (miles) **and** 305 (miles) **OR** No **AND** 69(.235) (litres) **OR** **No AND** 0.24(..) (lpm) **and** 0.22(7) (lpm) **OR** No **AND** 4.0(5..) (mpl) **and** 4.4(05..) (mpl)
Total marks for question			**4**		

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Question	Skills Standard	Process	Mark	Mark Grid	Evidence
Q9(a)	R3	Begins to find a route	1 or	N	Describes a route using at least 3 distances or 4 places, e.g. 3.1 + 4.3 + 4.2 (= 11.6) **OR** 8.1 + 4.2 + 4.8 (= 17.1) **OR** 5.1 + 5.5 + 4.3 (= 14.9) **OR** O to M to A to C **OR** O to A to C to M **OR** O to C to M to A
	A4	Identifies a complete route or finds the length of a complete route	2 or	NP	e.g. 3.1 + 4.3 + 4.2 + 4.8 (= 16.4) **OR** 5.1 + 5.5 + 4.3 + 3.4 (= 18.3) **OR** 8.1 + 4.3 + 5.5 + 4.8 (= 22.7) **OR** 3.1 + 5.5 + 4.2 + 3.4 (= 16.2) **OR** O to M to A to C to W **OR** O to C to M to A to W **OR** O to A to C to M to A to W **OR** O to M to C to A to W
	I7	Identifies shortest route and length of route	3	NPQ	O to M to C to A to W **and** 16.2 (miles) **OR** 3.1 + 5.5 + 4.2 + 3.4 = 16.2 (miles) May be indicated on diagram
Q9(b)	A5	Identifies another consideration when planning a route	1	R	Any valid consideration, e.g. road works, lots of traffic, speed limits, closed roads, traffic lights, one-way streets, time of day
		Total marks for question	**4**		







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